

Shading = zero extra service cost

Illustrative map - not to scale

Example 1: one station wins

Required regions: North, Central, South. Budget: 6.

Site	Activation cost	North	Central	South
A	3	0	0	9
B	4	9	0	0
C	6	0	0	0

North, Central, and South columns show extra service costs.

Open C. Assign all three regions to C.

Total = 6 + 0 + 0 + 0 = 6. Within budget: YES.

Conclusion: C is cheapest. Opening A and B would cost 7.

Example 2: two stations win

C now costs 8 to activate. The budget is now 7.

Site	Activation cost	North	Central	South
A	3	0	0	9
B	4	9	0	0
C	8	0	0	0

The service costs and required regions stay the same.

Open A and B. A serves North/Central. B serves South.

Total = 3 + 4 + 0 + 0 + 0 = 7. Within budget: YES.

Conclusion: A and B are cheapest. Opening only C costs 8.

Central can use either A or B at zero extra cost.

All costs use fictional units. Pay each open site's activation cost once.